

DAY 12 START — AUTO SCALING POLICIES + CLOUDWATCH ALARMS

Solomon, Day 11 DONE! You proved ALB → TG → ASG → Healthy!

Day 12 Mission: Make day5-asg SCALE automatically based on CPU!

TODAY'S 3 LABS:

LAB 1: Target Tracking Policy (Exam 60% of scaling questions)

ASG `day5-asg` Desired 1 Min 1 Max 3

Create policy: `cpu-50-target-tracking`

Metric: Average CPU 50%

This will auto add/remove `i-0bc45928f3e7b8efc` copies

LAB 2: CloudWatch Alarm + Step Scaling (Old but exam loves it)

Alarm: `day12-high-cpu` CPU > 70% for 2 mins

Alarm: `day12-low-cpu` CPU < 30%

Step policy: Add 1 if high, Remove 1 if low

LAB 3: Scheduled Action (SAA-C03 cost optimization)

Scale to 2 instances at 8am, back to 1 at 8pm (simulate business hours)

`day12-scale-out-morning` + `day12-scale-in-evening`

START NOW — Step 1:

1. Go EC2 → Auto Scaling Groups → day5-asg → Automatic scaling tab
2. Click Create dynamic scaling policy
3. Screenshot the Automatic scaling tab showing current policies (should be empty)

Save as `12D1-auto-scaling-empty.png`

Send that screenshot — then we create the CPU 50% target tracking policy!

SAA-C03 Tip for Day 12: Target Tracking creates 2 CloudWatch Alarms automatically (High + Low). Step scaling you create alarms manually. Exam always asks which to use → Answer: Target Tracking for CPU average!

DAY 12 FULL LECTURE — AUTO SCALING POLICIES (SAA-C03 CORE)

Complete Step-by-Step + 25 Exam Questions

PART 1: WHY WE NEED SCALING POLICIES?

Day 5 ASG you built is STATIC = Desired 1 always.

Real world traffic:

8am: 10,000 users → Need 3 instances

8pm: 100 users → Need 1 instance

Without policy: You manually change Desired 1→3 at 8am. You sleep, site crashes.

With policy: AWS does it automatically based on CPU, schedule, or prediction.

PART 2: 4 TYPES OF SCALING POLICIES (EXAM MUST KNOW)

1. TARGET TRACKING (80% of exam questions — YOU BUILT TODAY)

What: You say "Keep CPU at 50%" — AWS creates alarms and adds/removes instances to maintain 50%.

How it works:

You created `cpu-50-target-tracking` Target 50% Warmup 300

AWS auto-creates 2 CloudWatch Alarms:

`AlarmHigh` = CPU > ~60% → Add capacity

`AlarmLow` = CPU < ~40% → Remove capacity

Warmup 300 sec: New instance needs 300s to boot Apache before counting its CPU in average. Matches your ELB grace 300 + lifecycle 300 = BEST

PRACTICE!

When to use (Exam): Average CPU, Average RequestCountPerTarget, ALB Request count.

SAA Question Trick: If question says "keep CPU at 50% average" → Answer = Target Tracking. Never Step!

Your config proof:

Code

Policy: cpu-50-target-tracking

Type: Target tracking

Metric: Average CPU utilization at 50

Action: Add or remove capacity as required

Warmup: 300

2. STEP / SIMPLE SCALING (15% exam — Old but still asked)

What: You create CloudWatch alarm manually, then say what to do when alarm fires.

Example:

Alarm day7-high-cpu-70 CPU > 70% for 1 min → Policy: Add 1 instance

Alarm cpu < 30% → Policy: Remove 1 instance

Difference:

Simple: 1 step, waits cooldown before next scaling (old)

Step: Multiple steps based on how big breach is (e.g., CPU 70%→ Add 1, CPU 90%→ Add 3) — Better!

When to use: When you need custom steps or based on SQS queue length.

Cooldown: 300 sec default. Prevents flapping (scale out→scale in→scale out fast).

3. SCHEDULED SCALING (You BUILT TODAY — Cost Optimization!)

What: Scale based on TIME, not metric. For predictable patterns.

Your config:

scale-out-8am Recurrence 0 8 * * * Cron = 8am UTC daily Desired 2

scale-in-8pm Recurrence 0 20 * * * Cron = 8pm UTC daily Desired 1

Cron format: Minute Hour Day Month Weekday

0 8 * * * = Minute 0, Hour 8, every day/month/weekday = 8:00 daily

SAA Exam Trap:

Q: "Application has predictable traffic 9am-5pm high, night low. Cost optimize?"

Wrong: Target Tracking (wasteful, reacts after CPU high)

RIGHT: Scheduled Action 9am scale out, 5pm scale in ← Cheapest!

Start time in past error you got: AWS blocks if Start time < now. Fix = set tomorrow.

4. PREDICTIVE SCALING (5% exam — Newest)

What: Uses ML on past 14 days CloudWatch history to predict tomorrow's load and pre-scale BEFORE traffic hits.

Example: Your app gets 10k users every Monday 9am. Predictive learns pattern and adds instances at 8:50am Monday, BEFORE CPU spikes.

When to use: Cyclical, repeating patterns. Best combined with Target Tracking.

You see button: Create predictive scaling policy in your screenshot — That's it!

PART 3: STEP-BY-STEP YOU DID TODAY (Portfolio Steps)

Step 1: Target Tracking

1. EC2 → Auto Scaling Groups → day5-asg → Automatic scaling tab
2. Create dynamic scaling policy → Type Target tracking → Name cpu-50-target-tracking → Metric Avg CPU 50 → Warmup 300 → Create
3. Verify: Policy list shows 1, Details shows Enabled, Add/remove capacity as required
4. CloudWatch → Alarms → See 2 auto-created TargetTracking-...-AlarmHigh/Low

Step 2: Verify Alarms

Shows AWS auto-manages alarms. Don't manually delete those 2 alarms or policy breaks.

Step 3: Scheduled Actions

1. Same Automatic scaling tab → Scroll to Scheduled actions → Create scheduled action
2. scale-out-8am: Desired 2, Recurrence Every day Cron 0 8 * * *, UTC, Start tomorrow 08:00
3. scale-in-8pm: Desired 1, Recurrence Every day Cron 0 20 * * *, UTC, Start tomorrow 20:00
4. Verify: Scheduled actions (2) list shows both with Desired 2 and 1

PART 4: EXAM CHEAT SHEET — WHICH POLICY TO CHOOSE?

Scenario	Policy to Choose	Why
Keep CPU at 50% avg	Target Tracking	Exam default for avg metrics
SQS queue > 1000 msgs add 2 instances	Step scaling	Custom metric steps
Traffic high 9-5, low night	Scheduled	Predictable time-based = cheapest

Daily pattern repeating, want
pre-scale

Predictive + Target Tracking ML forecast

Need immediate scale on alarm

Step

Simple 1-step

Golden Rules:

1. Target Tracking creates its own CloudWatch alarms → Don't create separate alarm for CPU 50%
 2. Step/Simple requires YOU create alarm
 3. Scheduled needs at least Desired OR Min OR Max
 4. Warmup = Cooldown = Health check grace = 300 → Keep same to avoid race!
-

PART 5: 25 SAA-C03 EXAM QUESTIONS (Day 12 Focus)

Q1: ASG with Target Tracking CPU 50% currently 1 instance CPU 80% avg. What happens?

A: AlarmHigh fires, adds 1 instance, new desired 2, avg CPU drops toward 50.

Q2: Difference between Target Tracking and Step Scaling?

A: Target Tracking auto creates CW alarms and maintains target. Step requires manual alarms and custom adjustments.

Q3: App traffic predictable Mon-Fri 8am-6pm high. Cost optimize?

A: Scheduled actions scale-out 8am Desired 3, scale-in 6pm Desired 1. Cheaper than keeping 3 all day.

Q4: What is instance warmup 300 in target tracking?

A: Time new instance needs to boot before its metrics count toward average. Prevents premature scaling.

Q5: You have Target Tracking CPU 50%. CloudWatch shows AlarmLow In alarm CPU 20%. What happens?

A: Removes 1 instance (if min allows) to increase avg CPU back to 50.

Q6: Can you have multiple target tracking policies on same ASG?

A: Yes, but ASG will scale to max capacity required by any policy. Example CPU 50% needs 2, RequestCount 1000 needs 3 → ASG scales to 3.

Q7: Step scaling cooldown purpose?

A: Prevents flapping, waits 300s after scaling before allowing another scaling action.

Q8: Scheduled action recurrence `0 8 * * *` means?

A: Cron daily 8:00 AM UTC.

Q9: Why Target Tracking created 2 alarms?

A: One High breach, one Low breach to maintain target.

Q10: You set Desired 2 in scheduled action but ASG Max is 1. What happens?

A: Scheduled action fails, Desired cannot exceed Max. Increase Max to 3 first (you have Max 3).

Q11: Predictive scaling needs how much history?

A: 14 days minimum CloudWatch history to predict.

Q12: Best policy for ALB RequestCountPerTarget 1000?

A: Target Tracking RequestCountPerTarget 1000.

Q13: ASG Min1 Max5 Desired1. Scheduled action Desired 10 at 8am. What?

A: Fails if Max 5 < Desired 10. Must increase Max first.

Q14: You delete TargetTracking-...-AlarmHigh manually. What happens?

A: Policy breaks, no scale-out. Recreate policy.

Q15: Simple vs Step scaling difference?

A: Simple does one adjustment and waits cooldown. Step can do multiple adjustments based on breach size.

Q16: How to test target tracking quickly?

A: SSH to instance, run `stress --cpu 8 --timeout 300` to spike CPU >50%, watch new instance launch.

Q17: Scheduled action start time in past error fix?

A: Set Start date to tomorrow future time, not today 08:00 when now 17:33 UTC.

Q18: ASG has lifecycle hook Terminating:Wait 300s. How affects scaling?

A: When scaling in, instance goes Wait 300s to drain connections before terminate.
Matches warmup 300 best practice.

Q19: Cost: Keep 3 instances 24/7 vs Scheduled 1 night 3 day?

A: Scheduled cheaper saves 2 instances x 12 hours nightly.

Q20: Which policy for SQS queue depth scaling?

A: Step scaling based on ApproximateNumberOfMessages.

Q21: Target Tracking warmup vs Health check grace?

A: Both 300s best. Warmup = ignore new instance metrics 300s. Grace = don't health check new instance 300s.

Q22: You have 2 scheduled actions same time, one Desired2 one Desired4. Which wins?

A: Latest created wins (or larger Desired if overlapping). Avoid same time.

Q23: Can scheduled action change Min and Max too?

A: Yes, you can set Min, Max, Desired in same action.

Q24: What happens if both scale-out-8am and cpu-50-target-tracking want different capacity?

A: ASG takes max. If scheduled says 2 but CPU 80% needs 3, ASG goes to 3.

Q25: Final portfolio proof for Day12?

A: Screenshot showing: 1 target tracking policy cpu-50-target-tracking, 2 scheduled

actions scale-out-8am 2 / scale-in-8pm 1, 2 auto CW alarms TargetTracking-High/Low
OK, ASG At desired capacity 1/1 Healthy.